

Manual of ActFEMtide

Finite element simulation code for tide-induced electromagnetic field

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1 Introduction

ActFEMtide is a simulation code for calculating tide-generated electromagnetic fields via electromotive force in the ocean. The main solver is based on **ActFEM** ([Minami et al. 2018](#)) with the replacement of the source term of electric dipole source by tidally-induced electromotive force ($\mathbf{v} \times \mathbf{F}$). **ActFEMtide** is written in fortran and uses intel-math-kernel library (mkl) for the use of a sparse direct solver, PARDISO. ActFEMtide can use openMP for parallel computation with PARDISO, and MPI for parallel computation for multiple frequencies. **ActFEMtide** uses the meshgenerator developed for **TMTGEM** ([Minami et al. 2017](#)). ActFEMtide is composed of two parts: 1: Mesh generation part, 2: Simulation part. **ActFEMtide** is currently assume to use the output of tide model by ([Egbert and Erofeeva, 2002](#)). **ActFEMtide** uses [Gmsh](#) for mesh generation and the extrude algorithm described in [Oishi et al. \(2013\)](#), which was originally developed for **TMTGEM**. Most of the subroutines for extrusion are from a fluid simulation open-source code, [Fluidity](#).

Note that there are many copied contents from **TMTGEM** manual (v1.3).

2 Important notes

Coordinate system: X: eastward, Y: northward, Z: upward

Solved equations: Refer to [Minami et al. \(2018, EPS\)](#) (with modification to $\mathbf{v} \times \mathbf{F}$ source)

Input topography data: lon [deg], lat [deg], alt [m, **Downward Positive**]

Assumed tide model: [Egbert and Erofeeva \(2002\)](#)

3 Required environment

Please make the following packages installed in your PC:

- **Gmsh** (<http://gmsh.info/>) for mesh generation
- **Intel fortran compiler with mkl library** (confirm you can use “ifort -mkl=parallel ***”)
- **GMT** ([generic mapping tool](#)) for vieweing simulation results ghostscript (only for “gv” commands, e.g. in plot_z.sh in TMTGEM/Tohoku/flow)

If you can use Debian Linux distributions, “sudo apt install ***” help you install all the environments above.

4 Run sample codes in “Kikai/”

Run ActFEMtide anyway!! In the home folder of ActFEMtide/, the folder named Kikai/ is that for example simulations. To run the sample codes there are run by several steps.

Step0 Preparation of tide model

Compile the code ([Egbert and Erofeeva, 2002](#)) in ActFEMtide/mkfvxyz/ #####
\$cd mkfvxyz \$cd OTPS \$make (compile the codes for use of tide model)

If you are working in triton, copy data by the folloiwng command in ActFEMtide/mkfvxyz/ OTPS/ ##### \$.wget.sh Please confirm that in the DATA folder, the following files are prepared:

Step 1: “Preparation of topo” (ActFEMtide/Kikai/topo/)

convert etopo grd file to ascii *.xyz file.

```
$cd Kikai/topo
$./mk_etopo_kikai.sh          (etopo_kikai-1.xyz is generated)
```

Step 2: Generation of Tetrahedral mesh (ActFEMtide/Kikai/mesh/)

Generation of tetrahedral mesh

```
$cd ../mesh  
$./tetmeshgen.sh          (em3d.msh and others are generated)
```

It takes a while dependent on the spec of your PC. Many files are generated but the key three dimensional tetrahedral mesh file is “em3d.msh”. Please download the mesh file to local, and check it by the folloing step.

Step 3: Check the generated mesh files

After downloading the contents in ActFEM/Kikai/mesh

```
$gmsh em3d.msh
```

If you generate the mesh successfully, you can see the mesh.

Step 4: Preapre tidal flow and background magnetic field

```
$cd ../fvxyz  
$./mkfvxyz_mesh.sh
```

Step 5: run ActFEMtide

```
$cd ../fwd  
$./run_fwd.sh
```

Step 6: check the results

```
./plot_bxyz.sh  
./plot_ixyh.sh
```

You can geenrate two pdf files below.

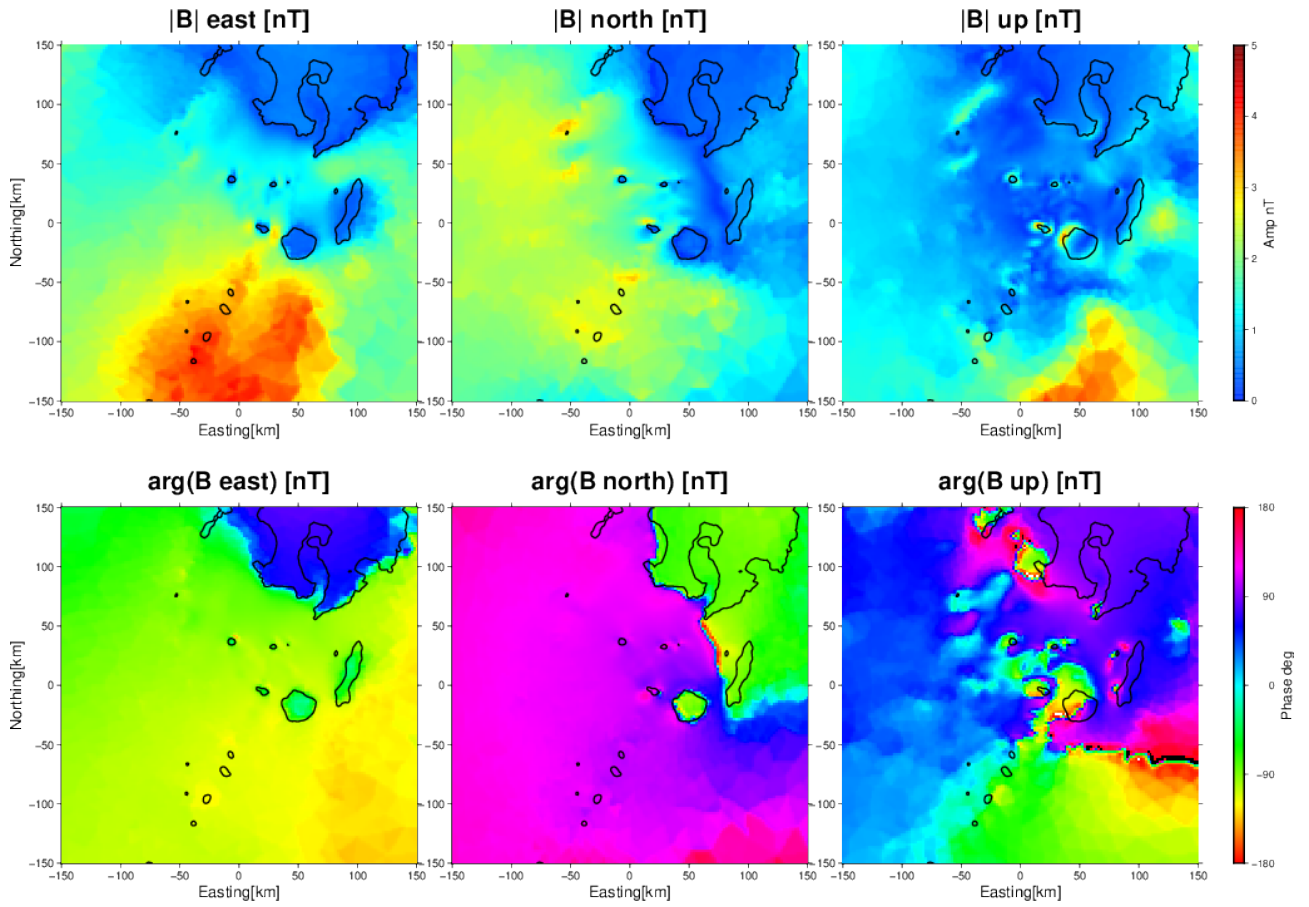


Figure 1. Tidally-induced magnetic fields due to M2 tide. All the three components are the values at the seafloor. The figure is generated by plot_bxyz.sh.

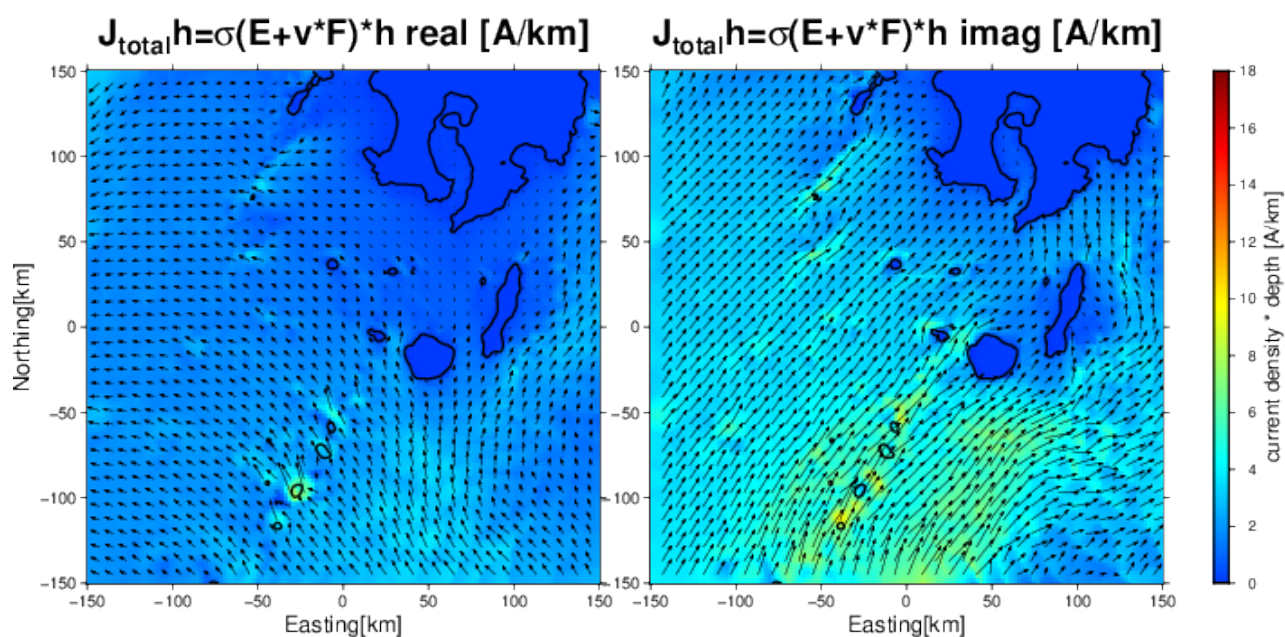
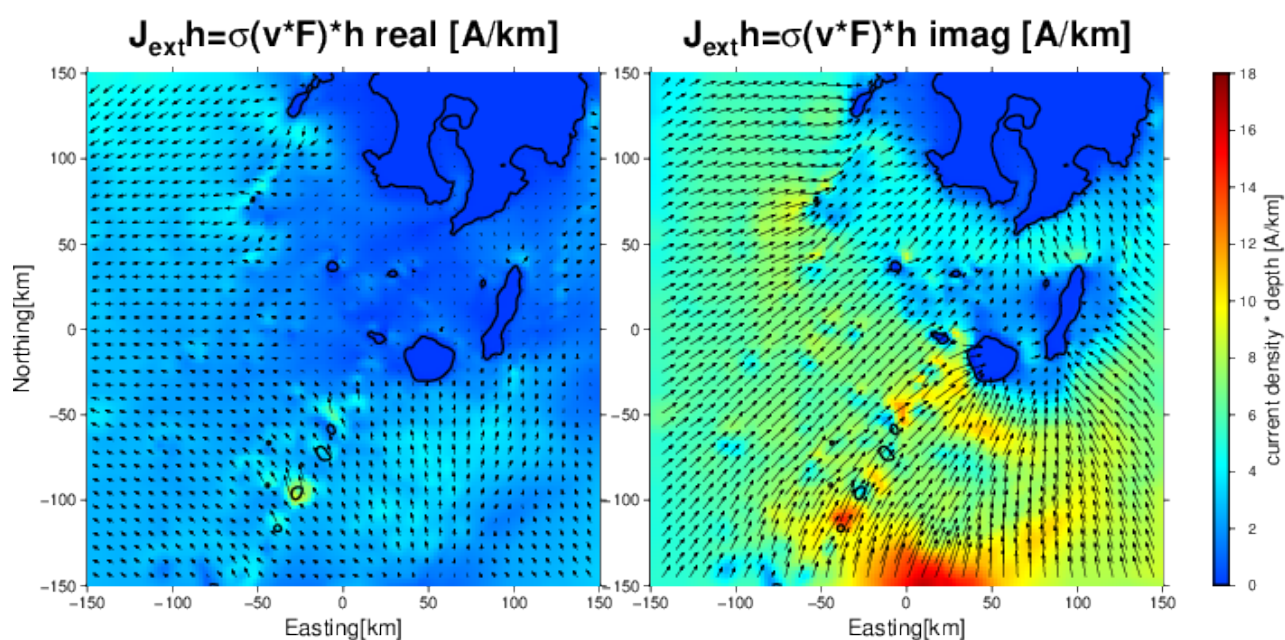
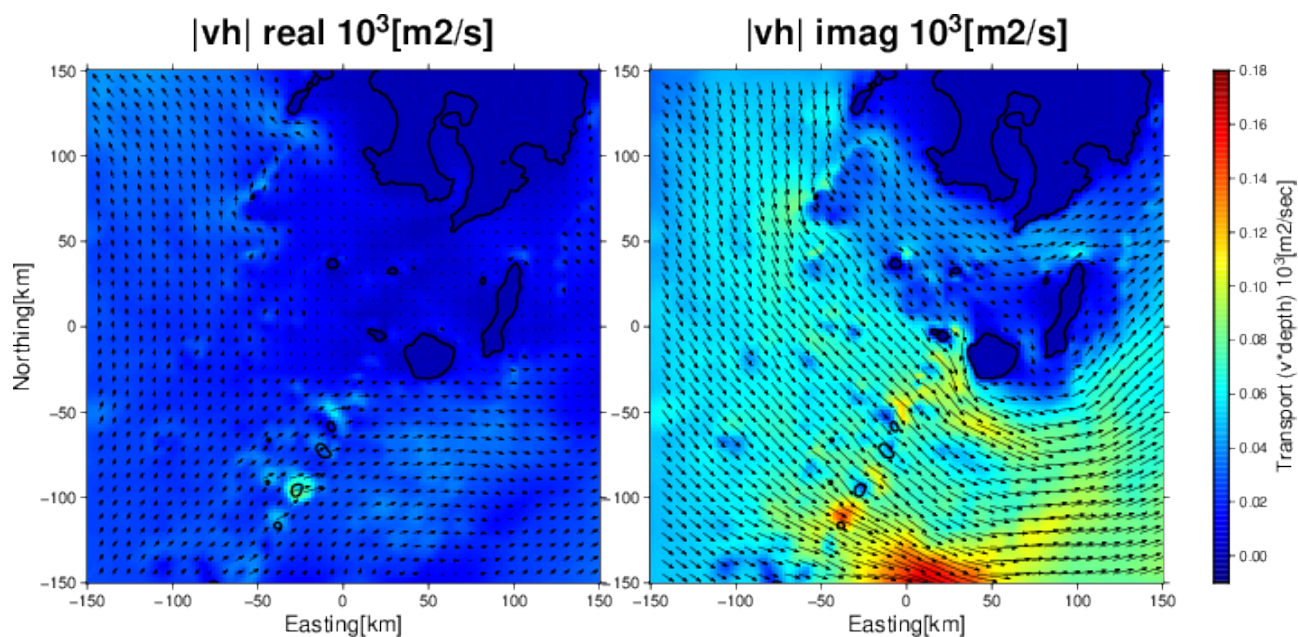


Figure 2. Real and Imaginary values of magnetic and electric fields due to M2 tide. The gifure is generated by plot_ixyh.sh.

References

- Egbert, G. D., & Erofeeva, S. Y. (2002). Efficient inverse modeling of barotropic ocean tides. *Journal of Atmospheric and Oceanic technology*, 19(2), 183-204.
- Minami, T., Utsugi, M., Utada, H., Kagiya, T., & Inoue, H. (2018). Temporal variation in the resistivity structure of the first Nakadake crater, Aso volcano, Japan, during the magmatic eruptions from November 2014 to May 2015, as inferred by the ACTIVE electromagnetic monitoring system. *Earth, Planets and Space*, 70(1), 138.
- Minami, T., Toh, H., Ichihara, H., & Kawashima, I. (2017). Three-Dimensional Time Domain Simulation of Tsunami-Generated Electromagnetic Fields: Application to the 2011 Tohoku Earthquake Tsunami. *Journal of Geophysical Research: Solid Earth*, 122(12), 9559-9579.